# ISA/IEC 62443-3-3 / 4-2 • SL-3 TESTING CHEAT SHEET

**Red Lion / N-Tron N-View (OPC Server & Diagnostic Monitor / N-View 2)** · Component type: **Software Application (SAR) - network monitoring / management software** · Target SL = SL 3

| | |
|---|---|
| **Where to work** | **N-View** on a Windows host: the **N-View OPC Server** + **Diagnostic Monitor** (per-port stats to OPC clients/HMIs); **N-View 2** adds SNMP auto-discovery + **centralized firmware management**. Switch configuration is done on the switch web UI (see the N-Tron 700 sheet) - N-View is monitoring + firmware push. Installer from Red Lion. |
| **Ports / paths** | N-View collects via the **N-View protocol** (Layer-2 multicast from N-View firmware) and **SNMP UDP 161** (N-View 2 discovery); serves data via **OPC DA (DCOM: RPC 135 + dynamic)** to OPC clients; firmware push via **TFTP/HTTP**. No TLS on the legacy OPC-DA/N-View path. Runs on Windows - harden the host. |
| **Notes** | SAR - Red Lion/N-Tron N-View monitoring software. It is largely **read-only monitoring**; the security-relevant write path is **N-View 2 firmware management** (it can push switch firmware - control who may and verify provenance). Legacy **OPC DA** (DCOM) + **SNMP** are the exposure. Confirm the N-View version; prefer SNMPv3 against the switches (configured on the switch). |
| **How to read** | Each row = native step-by-step (N-View / host) + command-level external-tool steps, noting where control is **delegated to the switch or host**. Default read-only. Mark Pass/Fail/Partial/N/A. Write/exploit tools = isolated env + written authorization. |

**Project capture & evidence handling - capture the full application backup + exports**

**Do this BEFORE working the checklist - screenshots of the config and programming screens are not enough on their own.** Capture the application's own backup and exports so nothing is missed:
**1. Take the native full backup / export first** (read-only, before anything): the N-View configuration / topology export, the managed-switch settings it holds, and the application logs.
**2. Export the security config:** users / roles / permissions, authentication / IdP settings, audit configuration, and TLS / certificate settings - as files and screenshots.
**3. Collect the logs:** application, service / wrapper, audit and event logs (exports), plus running-threads / diagnostics dumps where the product provides them.
**4. Capture identity & version:** product edition + version / patch level, installed modules / components, and the host OS / runtime (JRE / .NET) - screenshot the About / version screen.
**5. Name + protect (keep everything):** `<site>_<zone>_<assetTag>_<product>_<version>_<YYYYMMDD-HHMM>_<initials>` with the native backup extension + the exports; SHA-256 hash the backup; store **only** on the encrypted evidence share; the backup usually contains secrets / credentials - **never** email it in cleartext and follow the project retention / classification policy. Do not save admin passwords in tool profiles.
**6. Then analyze offsite** against the product-specific checklist, the product manuals, vendor research and experience; fill in the tool and auto-generate the report - and remember the manual fields the tool can't auto-write: the **executive-summary narrative**, the **overall assurance level + justification**, and the **punch-list**.

## FR 1 — Identification & Authentication Control (IAC)

| Req (3-3 / 4-2) | Requirement | Where & how to verify (native steps + Ext. tools) | Pass criteria / evidence | Notes / version |
|---|---|---|---|---|
| **SR/CR 1.1** | Human user identification & auth | N-View has **limited app-level login**; authentication is enforced (a) at the **host OS** - only authorized Windows users run N-View / the OPC server - and (b) at the **switch** - SNMP credentials (prefer v3) and the switch web login gate configuration. Verify both; record the N-View-level residual. Restrict the OPC server so only authorized OPC clients connect (DCOM permissions).<br>**Test:** confirm an unauthorized OPC client / Windows user cannot read the OPC server or push firmware.<br>**Ext. tools:** Host (who may run N-View); DCOM config (OPC client permissions); switch SNMPv3 creds (per switch sheet). | Host + switch enforce access; OPC/DCOM permissions restrict clients; N-View residual documented. | Partial/Compensating. |
| **SR/CR 1.2** | Software process / device identification | Confirm each monitored switch is uniquely identified in N-View (IP/MAC/name from discovery) and N-View monitors only the intended devices; SNMP engine IDs (v3) identify the switches.<br>**Ext. tools:** N-View device list; `nmap -sU -p 161 --script snmp-info <switch>`; Wireshark (N-View multicast / SNMP). | Unique switch identities in N-View; no rogue devices monitored. | |

| Req (3-3 / 4-2) | Requirement | Where & how to verify (native steps + Ext. tools) | Pass criteria / evidence | Notes / version |
|---|---|---|---|---|
| **SR/CR 1.3** | Account management | N-View has no user database - manage the host OS accounts that run it (least privilege), who may perform N-View 2 firmware pushes, and the switch SNMP/web accounts (per the switch sheet). No shared SNMP communities.<br>**Ext. tools:** Host accounts; N-View firmware-push authorization; switch SNMP/web accounts. | Host + switch accounts managed/least-privilege; firmware-push restricted. | Compensating. |
| **SR/CR 1.4** | Identifier management | Confirm unique host user IDs and unique switch identities (IP/SNMP engine ID); no reuse.<br>**Ext. tools:** Host/AD; N-View device list; inventory. | Unique host-user and switch identifiers; no reuse. | |
| **SR/CR 1.5** | Authenticator management | The authenticators are the switch SNMPv3 credentials + web password (strong, non-default - per the switch sheet) and the host/AD password policy. Avoid SNMP v1/v2 community strings; if legacy N-View requires v1/v2, treat the community as a secret + compensate with segmentation.<br>**Ext. tools:** Switch SNMPv3 config; host/AD policy; confirm v1/v2 communities not default. | Strong SNMPv3 + host credentials; defaults changed; v1/v2 avoided/compensated. | Partial (legacy SNMP). |
| **SR 1.6** | Wireless access management | **N/A.** Monitoring software with no wireless interface.<br>**Ext. tools:** Confirm no device wireless; host WLAN governed by OS/network. | N/A - justification recorded (software). | N/A |
| **SR/CR 1.7** | Strength of password-based authentication | Confirm the password-strength policy (length/complexity/lifetime) meets SL-3. **Steps:** for this product this is assessed with the same evidence as **SR/CR 1.5** above (shared steps); confirm this requirement individually and adjudicate it separately. | Strength of password-based authentication verified individually. | |
| **SR/CR 1.8** | Public key infrastructure (PKI) certificates | **N/A / limited.** Legacy OPC DA / N-View / SNMP have no TLS/PKI. Record N/A; provide confidentiality/integrity via segmentation and host controls; where SNMPv3 is used, its auth/priv (SHA/AES) is the crypto.<br>**Ext. tools:** Confirm no TLS on OPC-DA/N-View; SNMPv3 auth/priv (SHA/AES) in use. | N/A/limited PKI; SNMPv3 crypto where available; compensating segmentation. | N/A/Partial. |
| **SR/CR 1.9** | Strength of public key-based authentication | Confirm public-key authentication uses strong, current keys/algorithms. **Steps:** for this product this is assessed with the same evidence as **SR/CR 1.8** above (shared steps); confirm this requirement individually and adjudicate it separately. | Strength of public key-based authentication verified individually. | |
| **SR/CR 1.10** | Authenticator feedback | Confirm N-View / SNMP credential entry masks secrets (no cleartext echo); note SNMP v1/v2 community strings traverse the wire in cleartext -> use v3.<br>**Ext. tools:** Visual verification; Wireshark (confirm no cleartext v1/v2 communities on critical links). | Authenticators obscured; cleartext SNMP avoided (v3). | |
| **SR/CR 1.11** | Unsuccessful login attempts | Lockout is a switch/host concern, not N-View. The switch logs an **Authentication Error trap** on failed access - forward it; host/AD lockout for the N-View workstation.<br>**Ext. tools:** Switch SNMP Authentication-Error trap to the SIEM; host/AD lockout. | Auth failures trapped/forwarded; host lockout; N-View residual noted. | Delegated. |
| **SR/CR 1.12** | System use notification | No N-View banner; enforce the approved use notice on the host (GPO).<br>**Ext. tools:** Host `gpresult /h gp.html`. | Use notice at the host (compensating). | Compensating. |
| **SR 1.13** | Access via untrusted networks | Confirm the N-View host + switches are in the trusted OT/management zone; the OPC/DCOM + SNMP + N-View multicast are not exposed to untrusted networks; remote access via VPN/jump host.<br>**Ext. tools:** `nmap -Pn -p 135,161 <host/switch>` from the untrusted side (expect filtered); firewall review. | N-View/OPC/SNMP unreachable from untrusted networks; remote via VPN. | |

## FR 2 — Use Control (UC)

| Req (3-3 / 4-2) | Requirement | Where & how to verify (native steps + Ext. tools) | Pass criteria / evidence | Notes / version |
|---|---|---|---|---|
| **SR/CR 2.1** | Authorization enforcement | N-View is read-mostly; the write path is **N-View 2 firmware management** - confirm only authorized operators can push firmware (host account + N-View controls), and OPC clients get read-only data. The switch enforces config authorization (web/SNMP).<br>**Test:** as an unauthorized user, attempt an N-View 2 firmware push -> denied.<br>**Ext. tools:** N-View (firmware-push authorization); DCOM (OPC read-only); switch config rights. | Firmware push restricted; OPC read-only; switch config authorization enforced. | Partial/Delegated. |
| **SR/CR 2.2** | Wireless use control | **N/A.** Software, no wireless interface.<br>**Ext. tools:** Confirm no wireless. | N/A - justification recorded. | N/A |

| Req (3-3 / 4-2) | Requirement | Where & how to verify (native steps + Ext. tools) | Pass criteria / evidence | Notes / version |
|---|---|---|---|---|
| **SR 2.3** | Use control for portable/mobile devices | Control the N-View laptop and removable media (OS device-control); only authorized firmware images loaded for N-View 2 pushes; secure the host.<br>**Ext. tools:** Host removable-media GPO; firmware-image provenance. | Portable/engineering media controlled; firmware images verified. | |
| **SR/SAR 2.4** | Mobile code | No general-purpose mobile code; confirm only signed N-View runs and no untrusted add-ons; validate the installer.<br>**Ext. tools:** Installed-component review; verify installer signature. | Only signed/required components; no untrusted code. | SAR-specific. |
| **SR/CR 2.5** | Session lock | Confirm the host OS screen lock on inactivity; N-View re-auth via the host + switch SNMP creds.<br>**Ext. tools:** Host inactivity-lock GPO. | Host locks on inactivity. | Compensating. |
| **SR/CR 2.6** | Remote session termination | Confirm OPC client connections can be dropped and are bounded/monitored (DCOM); N-View 2 firmware sessions are single/controlled.<br>**Ext. tools:** OPC server session review; DCOM connection monitoring. | Sessions terminable; concurrency bounded/monitored. | |
| **SR/CR 2.7** | Concurrent session control | Confirm concurrent sessions are bounded/limited and monitored. **Steps:** for this product this is assessed with the same evidence as **SR/CR 2.6** above (shared steps); confirm this requirement individually and adjudicate it separately. | Concurrent session control verified individually. | |
| **SR/CR 2.8** | Auditable events | N-View has limited native audit; capture firmware-push actions on the host and forward switch **SNMP traps** (Authentication Error, Cold/Warm Start, Link Up/Down) to the SIEM; host audit for who ran N-View.<br>**Ext. tools:** Host audit (firmware pushes); switch SNMP traps -> SIEM. | Firmware actions + switch traps captured/forwarded; host attribution. | Compensating. |
| **SR/CR 2.9** | Audit storage capacity | Logging is off-device (host + SIEM) - confirm sizing/retention and alerting on collection failure.<br>**Ext. tools:** SIEM retention + source-silent alert. | Logs sized/offloaded; audit-failure alerting defined. | |
| **SR/CR 2.10** | Response to audit processing failures | Confirm a defined response to audit-processing/forwarding failures (alerting; no silent loss). **Steps:** for this product this is assessed with the same evidence as **SR/CR 2.9** above (shared steps); confirm this requirement individually and adjudicate it separately. | Response to audit processing failures verified individually. | |
| **SR/CR 2.11** | Timestamps | Confirm the host + switches are time-synced (NTP; switch SNTP) so trap/monitor timestamps correlate.<br>**Ext. tools:** `w32tm /query /status`; switch time (per switch sheet). | Accurate, synchronized timestamps. | |
| **SR/CR 2.12** | Non-repudiation | Confirm actions are attributable to a specific user (non-repudiation). **Steps:** for this product this is assessed with the same evidence as **SR/CR 2.8** above (shared steps); confirm this requirement individually and adjudicate it separately. | Non-repudiation verified individually. | |

## FR 3 — System Integrity (SI)

| Req (3-3 / 4-2) | Requirement | Where & how to verify (native steps + Ext. tools) | Pass criteria / evidence | Notes / version |
|---|---|---|---|---|
| **SR/CR 3.1** | Communication integrity | N-View OPC-DA / N-View multicast / SNMP v1/v2 are **unauthenticated cleartext** - contain them in a segmented management zone; use **SNMPv3** (auth+priv) where the switch supports it; do not expose OPC/DCOM across zone boundaries.<br>**Ext. tools:** Wireshark (OPC/N-View/SNMP; confirm in-zone + v3 where used); firewall review. | Monitoring comms contained/segmented; SNMPv3 where supported; no crossing the boundary. | Legacy - Compensating. |
| **SR/SAR 3.2** | Protection from malicious code (component) | Confirm N-View software integrity: install only the Red Lion-**signed** N-View (verify the installer signature / download from Red Lion); host anti-malware; restrict install to admins. Verify switch firmware images (for N-View 2 pushes) are Red Lion-provenance.<br>**Ext. tools:** `signtool verify /pa <setup>.exe`; host AV; firmware-image provenance. | Signed software validated; install restricted; firmware images verified. | SAR-specific. |
| **SR/CR 3.3** | Security functionality verification | After any change re-verify: DCOM/OPC client permissions, SNMPv3 use, firmware-push authorization, and patch level, against the hardening baseline; confirm the monitored switch security still matches its baseline (per the switch sheet).<br>**Ext. tools:** N-View/DCOM config review; switch config compare; version check. | Security-relevant settings match the baseline after change. | |

| Req (3-3 / 4-2) | Requirement | Where & how to verify (native steps + Ext. tools) | Pass criteria / evidence | Notes / version |
|---|---|---|---|---|
| **SR/CR 3.4** | Software & information integrity | Protect the N-View configuration/site data and any stored SNMP credentials/firmware images on the host: ACLs, encrypt at rest (BitLocker), back up under change control.<br>**Ext. tools:** Host ACL on N-View data; confirm no plaintext SNMP creds; backup review. | N-View data protected; credentials/images secured. | |
| **SR/CR 3.5** | Input validation | Patch known N-View / OPC CVEs; confirm N-View handles malformed switch responses without crashing.<br>**ISOLATED:** test against a bench switch.<br>**Ext. tools:** Nessus/OpenVAS on the host; Red Lion/CISA advisory cross-check; ISOLATED malformed-response test. | Known CVEs patched; malformed input handled safely. | |
| **SR/CR 3.6** | Deterministic Output | **N/A.** Device does not drive process outputs; deterministic/fail-safe output is not applicable. | N/A - justification recorded (not assessed as a control). | N/A |
| **SR/CR 3.7** | Error handling | Confirm N-View handles errors gracefully without leaking sensitive info (no plaintext SNMP community/credential in logs).<br>**Ext. tools:** N-View / host log review. | Graceful error handling; no credential leak. | |
| **SR/CR 3.8** | Session integrity | **N/A / limited.** Legacy OPC-DA/N-View/SNMPv1-2 sessions are not secured; SNMPv3 sessions are integrity-protected. Contain the rest by segmentation.<br>**Ext. tools:** Wireshark (SNMPv3 vs cleartext). | SNMPv3 sessions integrity-protected; legacy contained. | N/A/Partial. |
| **SR/CR 3.9** | Protection of audit information | Protect the host/SIEM audit stores; restrict who can read/clear them (host ACLs).<br>**Ext. tools:** Host ACL; SIEM access review. | Audit records protected from unauthorized read/modify/delete. | |

## FR 4 — Data Confidentiality (DC)

| Req (3-3 / 4-2) | Requirement | Where & how to verify (native steps + Ext. tools) | Pass criteria / evidence | Notes / version |
|---|---|---|---|---|
| **SR/CR 4.1** | Information confidentiality | Confirm information confidentiality — sensitive data is protected in transit and at rest. **Steps:** for this product this is assessed with the same evidence as **SR/CR 3.1** above (shared steps); confirm this requirement individually and adjudicate it separately. | Information confidentiality verified individually. | |
| **SR/CR 4.2** | Information persistence | On decommission/repurpose of the host, remove the N-View config/site data, stored SNMP credentials, firmware images, and backups; confirm no plaintext secrets persist; encrypt the host disk.<br>**Ext. tools:** grep for secrets; confirm data removed; BitLocker status. | N-View data/credentials purged on decommission; disk encrypted. | |
| **SR/CR 4.3** | Use of cryptography | N-View has no native transport crypto (OPC DA/N-View); use SNMPv3 (SHA/AES) against the switches and rely on host disk encryption + segmentation. Protect stored credentials.<br>**Ext. tools:** Confirm SNMPv3 auth/priv; host encryption; no plaintext creds. | SNMPv3 crypto where available; host encryption; compensating segmentation. | Partial. |

## FR 5 — Restricted Data Flow (RDF)

| Req (3-3 / 4-2) | Requirement | Where & how to verify (native steps + Ext. tools) | Pass criteria / evidence | Notes / version |
|---|---|---|---|---|
| **SR/CR 5.1** | Network segmentation | Confirm the N-View host + switches sit in a segmented management/OT zone behind a firewall; OPC/DCOM + SNMP + N-View multicast do not cross to IT; remote use via a jump host.<br>**Ext. tools:** `nmap -p 135,161 <host/switch>` per zone; firewall rule review. | Host + switches in a segmented zone; only required flows cross. | |
| **SR 5.2** | Zone boundary protection | Confirm zone-boundary protection — only required flows cross the boundary (system SR / network device).<br>**Steps:** for this product this is assessed with the same evidence as **SR/CR 5.1** above (shared steps); confirm this requirement individually and adjudicate it separately. | Zone boundary protection verified individually. | |

| Req (3-3 / 4-2) | Requirement | Where & how to verify (native steps + Ext. tools) | Pass criteria / evidence | Notes / version |
|---|---|---|---|---|
| **SR 5.3** | General-purpose person-to-person comms restrictions | **N/A.** Monitoring software with no general-purpose person-to-person messaging.<br>**Ext. tools:** Confirm no such services. | N/A - justification recorded. | N/A |
| **SR 5.4** | Application partitioning | Partition by criticality: separate N-View instances/zones and least-privilege host access; do not monitor unrelated criticality zones from one flat instance.<br>**Ext. tools:** N-View instance/zone segregation; host access review. | Monitoring scopes partitioned by criticality. | |

## FR 6 — Timely Response to Events (TRE)

| Req (3-3 / 4-2) | Requirement | Where & how to verify (native steps + Ext. tools) | Pass criteria / evidence | Notes / version |
|---|---|---|---|---|
| **SR/CR 6.1** | Audit log accessibility | This tool IS a monitoring layer: confirm N-View traffic/status + switch SNMP traps are readable/exportable and forwarded to the SIEM; N-View dashboards (or Crimson) provide continuous monitoring; passive OT IDS on the segment.<br>**Ext. tools:** N-View dashboard/export; switch traps -> SIEM; passive OT IDS. | Events accessible/exportable and monitored in near-real-time. | |
| **SR/CR 6.2** | Continuous monitoring | Confirm continuous monitoring — events forwarded to and monitored by a SIEM/OT-IDS in near-real-time. **Steps:** for this product this is assessed with the same evidence as **SR/CR 6.1** above (shared steps); confirm this requirement individually and adjudicate it separately. | Continuous monitoring verified individually. | |

## FR 7 — Resource Availability (RA)

| Req (3-3 / 4-2) | Requirement | Where & how to verify (native steps + Ext. tools) | Pass criteria / evidence | Notes / version |
|---|---|---|---|---|
| **SR/CR 7.1** | Denial of service protection | Confirm the N-View host is resourced/hardened and the OPC/SNMP polling load is bounded so a flood cannot deny monitoring; do not co-host unrelated apps; the N-View multicast load is contained to the management VLAN.<br>**Ext. tools:** Host resource monitor; polling-load review; ISOLATED load test. | Resources bounded; monitoring available under load. | |
| **SR/CR 7.2** | Resource management | Confirm resource management — resources are bounded; no exhaustion-based loss of function. **Steps:** for this product this is assessed with the same evidence as **SR/CR 7.1** above (shared steps); confirm this requirement individually and adjudicate it separately. | Resource management verified individually. | |
| **SR/CR 7.3** | Control system backup | Confirm the N-View config/site data + switch firmware images are backed up (versioned, offline) and a restore/rebuild is tested.<br>**Ext. tools:** Backup repository; restore test on a lab host. | Backups current and recovery validated. | |
| **SR/CR 7.4** | Control system recovery & reconstitution | Confirm recovery/reconstitution — a restore to a known-good state is tested. **Steps:** for this product this is assessed with the same evidence as **SR/CR 7.3** above (shared steps); confirm this requirement individually and adjudicate it separately. | Control system recovery & reconstitution verified individually. | |
| **SR 7.5** | Emergency Power | **N/A.** Emergency/backup power is provided at the facility/UPS level, not the device; confirm the device retains its security state across a power transfer. | N/A - justification recorded (not assessed as a control). | N/A |
| **SR/CR 7.6** | Network & security configuration settings | Harden: run N-View on a hardened least-privilege host, install only N-View, keep it patched, restrict DCOM/OPC to authorized clients, prefer SNMPv3 (disable v1/v2 on the switches), and remove sample data.<br>**Ext. tools:** Host/DCOM hardening review; N-View version; switch SNMP config. | Least functionality; DCOM/SNMP restricted; patched. | |
| **SR/CR 7.7** | Least functionality | Confirm least functionality — unused services/ports/protocols are disabled. **Steps:** for this product this is assessed with the same evidence as **SR/CR 7.6** above (shared steps); confirm this requirement individually and adjudicate it separately. | Least functionality verified individually. | |

| Req (3-3 / 4-2) | Requirement | Where & how to verify (native steps + Ext. tools) | Pass criteria / evidence | Notes / version |
|---|---|---|---|---|
| **SR/CR 7.8** | Component inventory | Inventory the N-View version, the host OS, and the monitored switches (model/firmware) so vulnerabilities track against Red Lion / CISA advisories.<br>**Ext. tools:** N-View version; host inventory; Red Lion/CISA feeds. | Accurate version + monitored-switch inventory maintained; advisories tracked. | |

## External Test Tools — how to use

| Tool / method | Verifies | How to use — steps / example | Constraints |
|---|---|---|---|
| **N-View (OPC Server + Diagnostic Monitor / N-View 2)** | Native monitoring & firmware review (all FRs) | Review the OPC server config (client/DCOM permissions), the discovered switch list, dashboards, and (N-View 2) firmware-management authorization. Switch config is on the switch sheet. | Read/authorized; firmware push authorized only. |
| **Host OS controls** | Who runs N-View + protect data (1.1, 2.1, 3.4, 4.2) | App-execution control, DCOM permissions (OPC clients), removable-media policy, ACLs + BitLocker on the N-View data. | Windows admin review. |
| **signtool** | Signed-installer validation (SAR 3.2) | `signtool verify /pa <setup>.exe`; download from Red Lion. | Read-only. |
| **nmap** | Switch/service exposure & segmentation (1.13, 5.x) | `nmap -sU -p 161 --script snmp-info <switch>`; `nmap -p 135 <host>`; scan per zone. | Read-only; authorized. |
| **Wireshark** | Comms integrity & SNMP (1.10, 3.1, 3.8) | Confirm N-View multicast/OPC/SNMP stay in-zone; SNMPv3 (not cleartext v1/v2) on management. | Passive; SPAN/tap only. |
| **Nessus / OpenVAS + Red Lion/CISA advisories** | CVE scan (3.5, 7.8) | Scan the N-View host for known CVEs; cross-check Red Lion / CISA advisories. | ISOLATED/representative; authorization. |

Controlled - Project Use · Independent Assurance Services · ISA/IEC 62443-3-3 / 4-2 · Product-specific companion to the 62443 assessment tool; confirm exact menu paths / CLI against the deployed firmware and the project hardening baseline. ISOLATED tools run only in a representative/lab environment with written authorization.